

FEDERAL MARITIME COMMISSION

Docket No. 17-09

FIAT CHRYSLER AUTOMOBILES NV, FCA US LLC, AND FCA ITALY S.P.A.

v.

**WALLENIOUS WILHELMSSEN LOGISTICS AS, WALLENIOUS WILHELMSSEN
LOGISTICS AMERICAS LLC, EUKOR CAR CARRIERS INC.,
NIPPON YUSEN KABUSHIKI KAISHA, NYK LINE (NORTH AMERICA) INC.,
MITSUI O.S.K. LINES, LTD., MOL (AMERICA) INC.,
KAWASAKI KISEN KAISHA, LTD., "K" LINE AMERICA, INC.,
COMPAÑÍA SUD AMERICAN DE VAPORES S.A., AND HÖEGH AUTOLINERS AS**

**MOTION *IN LIMINE* OF RESPONDENTS
WALLENIOUS WILHELMSSEN LOGISTICS AS AND
WALLENIOUS WILHELMSSEN LOGISTICS AMERICAS LLC
TO BAR COMPLAINANTS' APRIL 16, 2019 EXPERT REPORT**

Respondents Wallenius Wilhelmsen Logistics AS and Wallenius Wilhelmsen Logistics Americas LLC (collectively, "WWL"), by way of their motion *in limine* to bar the April 16, 2019 expert report served by complainants Fiat Chrysler Automobiles NV, FCA USA LLC, and FCA Italy S.p.A. (collectively, "complainants"), respectfully represent as follows:

BACKGROUND

1. Complainants originally filed this action against all respondents on or about October 6, 2017, which was docketed under No. 17-09.
2. On August 13, 2019, Administrative Law Judge Erin M. Wirth ("ALJ Wirth") entered a scheduling order that noted that, although the schedule requested by the parties "is longer than typically approved," the schedule was approved; that schedule provided, among

other things, that complainants' expert report(s) were to be served by no later than February 15, 2019.

3. Complainants failed to serve an expert report on or before February 15, 2019.

4. Five days later, on February 20, 2019, complainants moved to extend the August 13, 2018 scheduling order.

5. By an order dated February 26, 2019, ALJ Wirth granted in part and denied in part complainants' motion to extend the August 13, 2019 scheduling order. Specifically, the February 26, 2019 order provides that "[c]omplainants' expert witness report(s) [are] to be served" by April 16, 2019.

6. At 6:36 p.m. on April 16, 2019, complainants electronically served what they represented to be an expert report; a true and correct copy of the cover email and purported April 16, 2019 expert report of Christopher Rowland, Ph.D. ("the Rowland report") is attached hereto as **Exhibit "A"** and is made a part hereof by reference.

7. The Rowland report fails to meet the requirements of 46 C.F.R. § 502.141(d)(2) in every material respect.

8. As required by 46 C.F.R. § 502.71(a), on April 24, 2019, WWL conferred with complainants in an attempt to engage in a good faith effort to determine whether there is any opposition to the relief sought to this motion and, if there is opposition, to narrow the areas of disagreement. That attempt was unsuccessful.

9. This motion *in limine* followed.

**THE ROWLAND REPORT FAILS TO MEET THE REQUIREMENTS
OF 46 C.F.R. § 502.141(d)(2) IN EVERY MATERIAL RESPECT**

10. The requirements of 46 C.F.R. § 502.141(d)(2) are clear: every expert witness must produce a written report that contains:

- (i) A complete statement of all opinions the witness will express and the basis and reasons for them;
- (ii) The facts or data considered by the witness in forming them;
- (iii) Any exhibits that will be used to summarize or support them;
- (iv) The witness's qualification, including a list of all publications authored in the previous 10 years;
- (v) A list of all other proceedings or cases in which, during the previous 4 years, the witness testified as an expert in a trial, an administrative proceeding, or by deposition; and
- (vi) A statement of the compensation to be paid for the study and testimony in the proceeding.

The Rowland report meets none of the requirements of 46 C.F.R. § 502.141(d)(2).

11. **Description of the Rowland report.** The Rowland report consists of five parts. They are: (i) "Author and his Qualifications"; (ii) "Introduction and Assignment"; (iii) "Proposed Methodology"; (iv) "Profit Margins and Prices"; and (v) "Calculation of Damages".

12. **The Rowland report does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(i).** The Rowland report does not provide the "complete statement of all opinions the witness will express and the basis and reasons for them[.]" as required by 46 C.F.R. § 502.141(d)(2)(i). On the contrary, although the Rowland report purports to describe, in the most general terms possible, the methodology it intends to apply, no opinions at all are expressed and,

likewise, no reasons or bases for those opinions are presented.¹ Even assuming that the methodology proposed by the Rowland report is valid -- and it plainly is not -- there is no excuse for complainants' failure to timely complete and serve a valid expert report, particularly as the methodology relies on information that is and has been readily available to complainants. Therefore, the Rowland report does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(i).

13. **The Rowland report does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(ii).** The Rowland report also fails to set forth "[t]he facts or data considered by the witness in forming them[,]" as required by 46 C.F.R. § 502.141(d)(2)(ii). In fact, the Rowland report sets forth zero facts or data and, because no facts or data were considered, it does not tie the facts or data to any conclusions or opinions. Therefore, the Rowland report also does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(ii).

14. **The Rowland report does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(iii).** The Rowland report further fails to set forth "[a]ny exhibits that will be used to summarize or support [the opinions the report should express,]" as required by 46 C.F.R. § 502.141(d)(2)(iii). In fact, there are zero exhibits identified in the Rowland report, much less any exhibits listed as those that will be used to summarize or support the opinions the Rowland report should -- but most decidedly does not -- express. Therefore, the Rowland report also does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(iii).

¹ The methodology the Rowland report proposes to apply -- the "yardstick approach" -- is not appropriate; the "yardstick approach" is "employed to calculate lost profits by analyzing the sales and profits of comparable businesses." *Fairmont Supply Co. v. Hooks Indus.*, 177 S.W.23d 529, 532 (Tex. Ct. of App., First District 2005), *review denied*, 177 S.W.3d 529 (Tex. 2005) (emphasis supplied). Here, complainants do not seek lost profits, but reparations, that is, "payment . . . for actual injury caused by a violation" of the Shipping Act, 46 U.S.C. § 41305(b), and no informed person would credibly claim that the roll-on, roll-off ("RoRo") shipping industry and the container shipping industry are "comparable businesses," a core difference the Rowland report cavalierly elides.

15. **The Rowland report does not satisfy the requirements of 46 C.F.R. §**

502.141(d)(2)(iv). Additionally, the Rowland report further fails to set forth “[t]he witness’s qualification,² including a list of all publications authored in the previous 10 years[,]” as required by 46 C.F.R. § 502.141(d)(2)(iv). The *curriculum vitae* attached to the report lists four publications -- from 2009 to March 2017 -- none of which deal with ocean shipping, much less the RoRo shipment of vehicles. Instead, each of those publications is in the area of energy and energy demand. Therefore, the Rowland report also does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(iv).

16. **The Rowland report does not satisfy the requirements of 46 C.F.R. §**

502.141(d)(2)(v). Furthermore, the Rowland report further fails to set forth “[a] list of all other proceedings or cases in which, during the previous 4 years, the witness testified as an expert in a trial, an administrative proceeding, or by deposition[,]” as required by 46 C.F.R. § 502.141(d)(2)(v). The Rowland report makes no reference whatsoever of any proceedings or cases in which he testified as an expert in any proceeding. Therefore, the Rowland report also does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(v).

17. **The Rowland report does not satisfy the requirements of 46 C.F.R. §**

502.141(d)(2)(vi). Finally, the Rowland report further fails to set forth “[a] statement of the compensation to be paid for the study and testimony in the proceeding[,]” as required by 46

² Even the scant information provided by complainants demonstrates that complainants’ proposed “expert” is not qualified to opine on matters concerning the economics of the RoRo shipping industry; he compares two fundamentally and drastically non-comparable industries -- the RoRo shipping industry to the container shipping industry -- and his experience is not in any form of shipping -- much less the RoRo shipping industry -- but in energy. To the extent complainants’ proposed “expert” has any shipping knowledge or experience at all, it is limited to the liquid bulk shipping of energy products, that is, contract carriage, where the cargo owner owns or charters the vessel, with no independent cargo interests.

C.F.R. § 502.141(d)(2)(vi). Save for listing his hourly rate as \$385 per hour, and other than referring to “their normal and customary hourly rates[,]” there simply is no reference whatsoever to the compensation to be paid to his research staff or to his employer as a whole for the services provided and to be provided. Therefore, the Rowland report also does not satisfy the requirements of 46 C.F.R. § 502.141(d)(2)(vi).

18. **Discussion.** It is well-settled that “[t]he specific purpose of the [46 C.F.R. § 502.141(d)(2) expert report disclosure]³ requirement to exchange expert reports is to eliminate unfair surprise to the opposing party and to conserve resources.” *Headley v. Ferro Corp.*, 630 F. Supp. 2d 1261, 1266 (W.D. Wash. 2008) (citation omitted).

19. In practice, that means that an expert’s report “must set forth facts and, in doing so, outline a line of reasoning arising from a logical foundation.” *R.C. Olmstead, Inc. v. CU Interface, LLC*, 606 F.3d 262, 271 (6th Cir.), *reh’g en banc denied*, 210 U.S. App. LEXIS 13117 (6th Cir. June 23, 2010) (citation, internal quotation marks and editing marks omitted).

20. In considering complainants’ failure to comply with the requirements of 46 C.F.R. § 502.141(d)(2) and the February 26, 2019 order that “[c]omplainants’ expert witness report(s) [are] to be served” by April 16, 2019, an overarching notion governs: “Obedience to the constraints of the Scheduling Orders of the Court is critical, if the Court is to capably perform its case management responsibilities. Accordingly, the flouting of discovery deadlines cause substantial harm to the judicial system.” *Upsher-Smith Labs. v. Mylan Labs.*, 944 F. Supp., 1411, 1438 (D. Minn. 1996) (citation, internal quotation marks and editing marks omitted). That is so because the required expert disclosures are designed to insure that “opposing parties have a

³ The requirements of 46 C.F.R. § 502.141(d)(2) and Fed. R. Civ. P. 26(a)(2)(B) are, in all fundamental respects, identical. For that reason, case law interpreting what is required under Fed. R. Civ. P. 26(a)(2)(B) is cited here in support of this motion.

reasonable opportunity to prepare for effective cross examination and perhaps arrange for expert testimony from other witnesses.’” *Id.* at 1439-40 (quoting Advisory Committee Notes to the 1993 Amendments to Rule 26(a)(2)(B)). And, “as a sanction for a party’s failure to comply with the requirements for expert disclosure, a Court may exclude the testimony of that expert.” *Id.* at 1440 (citations omitted).

21. Facing like circumstances, in an action seeking to recover ERISA benefits, then-U.S. District Judge (now U.S. Circuit Judge) Jordan struck an expert report and excluded that proposed expert’s testimony, explaining:

I agree that the Report fails to even come close to satisfying the requirements for expert reports under Federal Rule of Civil Procedure 26(a)(2)(B). Specifically, the Report does not contain [the proposed expert]’s qualifications to give expert testimony, namely an identification of his employment history, or identification of publications, or a listing of other cases in which he has testified, or an indication of his compensation from Plaintiffs. [The proposed expert] states in the Report that “estimates will be made of the pension [plaintiff] would have received under the Consol Pension Plan ... [.]” but does not provide any indication of those estimates or what his conclusions based on those estimates will be. [The proposed expert] chooses . . . wage growth rates for a number of years without any explanation of how or why he chose those numbers. Thus, the Report fails to contain a complete statement of all opinions to be expressed and the basis and reasons therefor as required by Rule 26(a)(2)(B).

Moreover, [the proposed expert]’s proposed testimony and Report fail to satisfy the *Daubert* [*v. Merrell Dow Pharms., Inc.*, 509 U.S. 579 (1993)] standards for the admissibility of expert testimony. Without including [the proposed expert]’s qualifications, there is no way for me to determine whether he is qualified to render the expert opinions he has proposed in the Report. There are also clear analytical gaps in [the proposed expert]’s opinions, namely, the methodology he used. . . . It is therefore impossible to judge the reliability of [the proposed expert]’s methods. Thus, [the proposed expert]’s report fails to meet the requirements of both Rule 26(a)(2)(B) and *Daubert*.

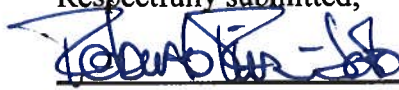
[*Pell v. E.I. DuPont de Nemours & Co.*, 231 F.R.D. 186, 193-94 (D. Del. 2005) (parentheticals omitted).]

CONCLUSION

22. Complainants could not have served the Rowland report with any reasonable or honest expectation that it satisfied the clear-cut requirements of 46 C.F.R. § 502.141(d)(2). It is obvious that complainants served the Rowland report for a different reason: solely as a place-keeper, a device to allow complainants to make the empty claim that they satisfied the requirements of the February 26, 2019 order and 46 C.F.R. § 502.141(d)(2) when, in truth and in fact, all complainants served was a less-than-barebones “report” that did not address any of the requirements of the February 26, 2019 order and 46 C.F.R. § 502.141(d)(2). That bit of legal legerdemain reeks of contempt for the processes by which these proceedings are to be conducted.

WHEREFORE, for the foregoing authority, arguments and reasons, WWL respectfully requests that the Rowland report be stricken and that complainants be held to have forfeited any ability to submit an expert report or to present expert testimony in these proceedings.

Respectfully submitted,



Roberto A. Rivera-Soto

Jason A. Leckerman

BALLARD SPAHR LLP

210 Lake Drive East – Suite 200

Cherry Hill, New Jersey 08002-1163

Telephone: (856) 761-3400

riverasotor@ballardspahr.com

leckermanj@ballardspahr.com

Paul M. Heylman

SAUL EWING ARNSTEIN & LEHR LLP

1919 Pennsylvania Av., NW. - Suite 550

Washington, D.C. 20006-3434
Telephone: (202) 343-3422
paul.heyman@saul.com

*Counsel for respondents Wallenius Wilhelmsen
Logistics AS and Wallenius Wilhelmsen Logistics
America LLC*

DATED: April 24, 2019

CERTIFICATE OF SERVICE.

I hereby certify that, on April 24, 2019, true and correct copies of the foregoing motion *in limine* was filed electronically with the Secretary of the Federal Maritime Commission, and electronic copies thereof were sent to counsel for complainants in Docket No. 17-09.

DATED: April 24, 2019



Roberto A. Rivera-Soto

EXHIBIT “A”

Rivera-Soto, Roberto A. (NJ)

From: Sprovtsoff, Jessica <JSprovtsoff@schiffhardin.com>
Sent: Tuesday, April 16, 2019 6:37 PM
To: Rivera-Soto, Roberto A. (NJ); 'jlawrence@cozen.com'; 'Rohde, Wayne'; 'Melissa H. Maxman'; Leckerman, Jason (Phila); 'Heylman, Paul M. (Paul.Heylman@saul.com)'; 'Antonelli, Matthew'
Cc: Wierenga, Robert J.
Subject: Fiat Chrysler Automobiles NV, et al. v. Wallenius Wilhelmsen Logistics AS, et al., FMC Docket No. 17-09
Attachments: Expert Report of Christopher Rowland PhD.PDF

⚠ EXTERNAL

Please see the attached expert report. We anticipate supplementing (and reserve the right to do so) this report as we obtain additional discovery and information from other sources.

Have a good evening,
Jessica

Jessica Sprovtsoff

Partner
jsprovtsoff@schiffhardin.com
d +1.734.222.1518 | f +1.734.222.1501

Schiff Hardin LLP
350 South Main Street | Suite 210 | Ann Arbor, MI 48104
schiffhardin.com | v-card | view bio

This message and any attachments may contain confidential information protected by the attorney-client or other privilege. If you believe that it has been sent to you in error, please reply to the sender that you received the message in error. Then delete it. Thank you.

BEFORE THE FEDERAL MARITIME COMMISSION

**CONFIDENTIAL--TO BE FILED UNDER SEAL
SUBJECT TO PROTECTIVE ORDER**

**FIAT CHRYSLER AUTOMOBILES NV,
FCA US LLC and FCA Italy S.p.A.,**

Complainants,

No. 1234937

v.

**Wallenius Wilhelmsen Logistics AS,
Wallenius Wilhelmsen Logistics Americas
LLC, EUKOR Car Carriers Inc., Nippon
Yusen Kabushiki Kaisha, NYK Line (North
America) Inc., Mitsui O.S.K. Lines, Ltd.,
MOL (America) Inc., Kawasaki Kisen
Kaisha, Ltd., "K" Line America, Inc.,
Compania Sud Americana de Vapores, and
Hoegh Autoliners AS**

Respondents.

EXPERT REPORT OF CHRISTOPHER ROWLAND, PH.D.

Econ ONE Research, Inc.

April 16, 2019

Suite 1025
1 Greenway Plaza
Houston, Texas 77046

TABLE OF CONTENTS

I.	Author and Qualifications.....	1
II.	Introduction and Assignment.....	1
III.	Proposed Methodology	2
IV.	Profit Margins and Prices	3
V.	Calculation of Damages.....	4

I. Author and Qualifications

1. My name is Dr. Christopher Rowland. I am a Senior Economist with Econ One Research, Inc., an economic research and consulting firm headquartered in Los Angeles, California with offices in Houston, Texas and in a number of other cities. I have a master's degree in industrial engineering and a doctoral degree in economics, both from Texas A&M University and a bachelor's degree in economics from California State University, Chico. My doctoral work concentrated on the field within economics known as industrial organization, which involves among other things the study of markets, competition, antitrust, and other forms of regulation.
2. I have over a decade of experience conducting economic research and analysis. I have three years of experience with shipping economics and logistics as I designed and implemented a global shipping model for Chevron. This work entailed accounting for shipping economics and logistics for the purposes of predicting routing, fuel consumption, and capacity changes for the vast majority of the global deep water fleet. My biographical information is included as Appendix 1.
3. Econ One is being compensated for the time I spend on this matter at my normal and customary rate of \$385 per hour. Econ One also is being compensated for time spent by my research staff on this matter at their normal and customary hourly rates.

II. Introduction and Assignment

4. I have been informed by Schiff Hardin ["Counsel"] that I can assume that the reader has read the Statement of Claim and is familiar with the parties, concepts, facts and allegations to which are referred therein. For convenience, I have adopted within my Report the definitions used in the Statement of Claim.
5. I have been retained on this case to serve as an economic expert on matters of overcharges incurred by FCA as a result of challenged conduct by shipping companies operating Roll On/Roll Off ["roro"] vessels from the period spanning 2014 to present. I have been instructed by counsel to assume that the challenged conduct occurred, leading to higher prices. I have been asked to propose a feasible and reliable methodology for estimating damages resulting from overcharge. I have also been asked to describe how this methodology would arrive at a damages estimate that is the result of illicit challenged behavior. Herein I propose a common economic

methodology utilizing data that can reasonably be expected to become available through continuing research, the client, and the discovery process. I further offer my opinion as to why this methodology would produce reliable estimates of the damages that resulted from the alleged collusion.

6. My staff and I have reviewed a variety of materials in undertaking this assignment, including documents provided by FCA ["Claimants"], and publicly available information. Appendix 2 sets forth a list of this information. I also have relied upon my professional training and experience as an economist. I understand that as this matter proceeds the parties may agree or be ordered to produce further documents in due course. To the extent that additional information becomes available through that process that bears on my opinions, I expect to review that information and supplement my opinions and preferred methodology as appropriate.

III. Proposed Methodology

7. In my opinion it is possible to reliably estimate overcharges incurred by FCA using a common economic procedure known as a "yardstick approach", which I will now describe. First, an industry that is both similar, and which did not engage in illicit collusion is selected for comparison as a source of more competitive profit margins and prices. Then the components of profit margins are collected from the two industries (in this instance those components are revenue and Earnings Before Interest Taxes Depreciation and Ammortization ["EBITDA"]).
8. Next, factors unrelated to illicit collusion, but which might lead to differences between the margins of the two industries are considered. These factors are used to adjust profit margins so that margins in the two industries can be considered on a consistent basis. After making these adjustments, the remaining difference in margins between the two industries represents the overcharge resulting from collusion. This is because other factors that may explain the difference have already been taken into account, such that any differences that remain can be attributed to collusion.
9. For estimating damages from overcharges, I would calculate the percentage change in roro freight rates required to cause margins to be identical in both industries over the challenged period. This percentage change represents the percentage of overcharge

in ro-ro freight rates. I would then multiply the total freight charged to FCA over the challenged period by the percentage of overcharge to arrive at damages.

10. The industry I would use for comparison is the containership industry. The reasons I find this to be an appropriate industry for comparison is that it satisfies both of the criteria that I stipulated for comparison, being simultaneously more competitive¹ and also similar in many ways to the ro-ro industry. These industries are similar because both involve large international firms engaged in international shipping operations with large and geographically diversified fleets. Both operate extremely large vessels with similarly trained crews, use similar fuels, and share many of the same ports and routes. On occasion both industries even transport the same cargo (vehicles). For these reasons containerships make an appropriate candidate for comparison.

IV. Profit Margins and Prices

11. The approach I described entails comparing profit margins, specifically the EBIDTA margin, which is calculated as the simple ratio of EBIDTA/revenue. I will now discuss the reasons for my preference in this matter towards the comparison of profit margins as opposed to the comparison of prices.
12. Profit margins in an industry represent the return an investor could hope to achieve by investing in that industry, all things constant. Two similar industries will then attract the same grouping of investors and investment dollars. If one of those industries offers consistently higher margins, investment dollars will begin moving towards the higher margin industry and away from the lower margin industry. Those investments will manifest as increased production or supply, heightening competition and lowering prices and margins. This behavior will continue until margins equilibrate between the two industries.
13. Unlike margins, prices do not necessarily have this same equilibrating force even amongst similar industries if the units they represent are not identical. The reason is that prices per unit depend on, among other things, the unit or good being sold. For instance, even though both industries are selling transportation services for freight,

¹ Goulielmos, Alexander M. 2017. "Containership Markets: A Comparison with Bulk Shipping and a Proposed Oligopoly Model." *SPOUDAI - Journal of Economics and Business*, Vol. 67, No. 2, pp. 47-68.

the units in the roro industry are Car Equivalent Units ["CEUs"], whereas in the containership industry they are Twenty foot Equivalent Units ["TEUs"]. The former is a vehicle, and the latter is virtually any commodity a customer could pack into a standard shipping container measuring 20 feet in length. Since these units represent fundamentally different things, there is no economic mechanism working to equilibrate their prices. Profit margins on the other hand are expressed as percentages and represent the same thing, return from investment, making their direct comparison appropriate.

V. Calculation of Damages

14. The economic task associated with measuring overcharges resulting from a conspiracy of the sort alleged in this case is to isolate the effects on margins and ultimately prices that are attributable to collusion as opposed to innate differences between industries or distinct changes in supply and demand. Once these factors are taken into account, the remaining deviation between margins in the two industries is attributable to collusion.
15. To begin accounting for innate differences like routing geography I would select containership companies that operate on similar routes, or routes of similar distance, to the roro provider whose overcharge is being estimated. Other innate differences will be used to directly modify margins. For instance, fuel consumption rates may be different due to differences in ship design. Fuel cost is born by the shipper in most instances, such that the shipper must recoup this cost through the freight rate that they charge for the route. For a given shipping company (either container or roro) this is addressed by deducting their reported fuel cost from their total freight revenue. As freight revenue is a component of the EBIDTA margin, this would remove the impact of differences in fuel costs from both margin calculations, making the two margins more comparable. This same technique is to be used for other innate differences.
16. Market conditions must also be considered. Market conditions that might affect prices and margins and induce disparities between the two industries would likely include: differences in excess capacity of the fleets, freight demand, demand trajectory, new ship additions, and retirements of older vessels. These factors must also be considered in determining the direction and magnitude of the adjustment

between the two sets of margins. Thus, the analysis and calculations will focus on periods where market conditions are similar.

17. Once margins between roro and container fleets have been adjusted to account for both innate differences and differences in market factors, the percentage difference between the two sets of margins over the challenged period is interpreted as being due to overcharges arising from the illicit collusion. The total damages estimate is then calculated by multiplying this overcharges estimate by the total freight cost incurred by FCA over the challenged period.



Christopher Rowland, Ph.D.
April 16, 2019



Chris Rowland
Senior Economist
Houston, Texas
Tel: 713 228 2700

EDUCATION

Doctorate: Applied Economics, Texas A&M University College Station, TX
Specialization: Industrial Organization, Energy, Policy, and Econometrics

Master's degree: Industrial Engineering, MEng, Texas A&M University College Station

Bachelor's degree: Bachelor's in Economics (minor in mathematics), BA, California State University Chico, CA

RESEARCH AREAS

Oil and Gas, Refining, Commodity Markets, Derivatives, Policy Analysis, Demand System Analysis, Demand forecasting, Supply Modeling, Linear Programming, Time Series, Forecasting.

PUBLICATIONS

Policy Implications of Considering Pre-commitments in US Aggregate Energy Demand System - *Energy Policy* V. 102, March 2017, pages 406-413

Politics and Petroleum: Unintended Implications of Global Oil Demand Reduction Policies- *Energy Research and Social Science* V. 11, January 2016, pages 209-224

Optimal Generation Technology Given Electricity Storage in Electricity Markets - 2010

Modeling World Crude Oil Supply by Reservoir Portfolio - 2009

WORK EXPERIENCE

Econ One

Senior Economist 2018-Present

- Economic consulting specializing in energy, industry, and transportation
- Provide expert opinion and reporting on economic matters

Chevron

Senior Commercial Advisor, Crude Supply and Trading 2017-2018

- Design, implement, and test trading algorithms for crude oil and derivatives
- Develop commodity trading strategies
- Guide crude oil pricing strategies for equity production
- Executed both physical and paper trades for Middle East crudes

Chevron

Senior Energy Economist, 2015 - 2017

- Designed and created new demand model to simultaneously forecast demand for multiple refined products and energy commodities with a much higher degree of accuracy than previous tools.
- Generate demand forecast models for distillates, gasoline, fuel oil and natural gas.
- Consult for crude trading group to develop strategic plays and methods
- Creation and management of Chevron's global shipping model - The development of this model was a major cross functional endeavor involving numerous subject matter experts. The insights provided from this model have been instrumental in forming Chevron's strategy for responding to the imposition of the 2020 IMO sulfur cap.
- Coordinate analytical work and oversee model development of Chevron's three strategy and planning groups that jointly forecast demand, supply, and prices.
- Regularly design and deliver presentations for top executives and joint venture partners on the global refining outlook and pressing issues such as current and expected developments in bunker fuel demand and personal mobility.

Chevron

Fundamental Demand Analyst Distillates, 2013 - 2015

- Forecasting diesel and jet fuel demand for 32 regions and globally. Forecasts used in both regional analysis and for global product and crude price forecasting.
- Assist with short term planning and forecasts for Value Chain Optimization, Product Supply and Trading, and Area Strategic Plan for refinery optimization.
- Maintain expertise on energy policies, technology and mobility trends around the world.

Zajonc Corporation

Lead Evaluator, 2011 - 2013

- Lead several evaluation projects which involved establishing timelines and deliverables for my colleagues, communicating with grant managers/stakeholders and reviewing/compiling work.
- Consulted grant manager on budget keeping and reporting.
- Actively acquired new projects by writing/assisting in application process.
- Suggested and built a model to demonstrate the feasibility of a new business offering that would increase operating revenues by 30% and help to vertically integrate the business.

Texas A&M University

Research/Teaching Assistant, 2007 - 2010

- Project modeling ethanol production and supply chain in Texas.
- Review papers, mine data and research modeling methods for journal submission.
- Lecture, coordinate and teach bi-weekly review sessions.

Royal Dutch Shell plc.:

Innovation Analyst for Gamechanger group, 2006 – 2007

- Research and data analysis with regard to emerging technologies and business trends.
- Consulting on merit and economic viability of proposed investment decisions.
- Monitored evolving trends in mobility and technology such as battery development and electric vehicles.
- Generating reports for senior management and decision makers.

COMPETENCES AND SKILLS

Palisades software (@risk, decision tree, top rank) MS Office (including Visual Basic for Excel, Access and SQL), Matlab, SAP, Stata, SAS, R, and Python

Appendix 2
In FCA v. WWL et al. Antitrust Litigation
List of Materials Reviewed

Includes all documents, studies, and articles cited in the Report.

Complaint, *FCA v. WWL et al.*, October 6, 2017.

Documents

FCA Transit Times.xlsx
Fuel Program Calculations for Ocean Carriers
Global Tender Summary_2009.xlsx
Global Tender Summary_2014.xlsx
Hapag-Lloyd Automotive Presentation Feb 2017
HIS Markit Global Production by Region & Sales Region (2009-2019CY).xlsx
Index of Letters and Contracts Between FCA and Ocean Shippers.xlsx
Ocean PPT
Ocean Spend 2010-2017.xlsx
PPT for FCA (July 2018)
Product Committee Meeting Minutes - Final (October 11, 2011)
RORO Industry August 2018
Standardised Bunker Clause Chrysler (Exhibit A)
Undercoating Analysis

Publicly Available Materials

Alphaliner. "Alphaliner - The Worldwide Reference in Liner Shipping." <https://www.alphaliner.com/>.
Alphaliner. 2016. "Alphaliner Monthly Monitor - September 2016."
American Fuel & Petrochemical Manufacturers. 2017. "AFPM United States Refining and Storage Capacity Report."
Croatian Shipbuilding Corporation. 2012. "Ferries & Ro-Ro, Yachts, High Speed Boats." <http://hb.hr/wp-content/uploads/2014/12/ferries.pdf>.
DHL. "DHL Global." <http://www.dhl.com/>.
DHL. 2019. "Ocean Freight Market Update."
Goulielmos, Alexander M. 2017. "Containership Markets: A Comparison with Bulk Shipping and a Proposed Oligopoly Model." *SPOUDA* - *Journal of Economics and Business*, Vol. 67, No. 2.
Maersk. "APMM Quarterly Figures Q1 2010 to Q4 2014."
Maersk. "Maersk - The World's Largest Container Shipping Company." <https://www.maersk.com/>.
Maersk. "Maersk Quarterly Figures Q1 2010 to Q4 2015."
Maersk. 2014. "Annual Report."
Maersk. 2014. "Presentation Annual Report."
Maersk. 2015. "Annual Report."
Maersk. 2015. "Presentation Annual Report."
Maersk. 2016. "Annual Report."
Maersk. 2016. "Presentation Annual Report."
Maersk. 2017. "Annual Report."
Maersk. 2017. "Annual Report."
Maersk. 2017. "Presentation Annual Report."
Maersk. 2018. "Annual Report."
Maersk. 2018. "Presentation Annual Report."
McCrary, Justin, and Daniel L. Rubinfeld. 2014. "Measuring Benchmark Damages in Antitrust Litigation." *Journal of Econometric Methods*, Vol. 3, No. 1.
Mitsui O.S.K. Lines. "Mitsui O.S.K. Lines Home." <https://www.mol.co.jp/en>.
United Nations Conference on Trade and Development. "UNCTAD Home." <https://unctad.org/en/Pages/Home.aspx>.
United Nations Conference on Trade and Development. 2017. "Review of Maritime Transport."
United Nations Conference on Trade and Development. 2018. "Fact Sheet #14: Merchant Fleet." *UNCTAD Handbook of Statistics*.
United Nations Conference on Trade and Development. 2018. "Fact Sheet #15: Maritime Transport Indicators." *UNCTAD Handbook of Statistics*.
Wallenius Wilhelmsen. "Wallenius Wilhelmsen Logistics ASA." <https://www.walleniuswilhelmsen.com/>.
Wilh. Wilhelmsen ASA. 2015. "Quarterly Report: First Quarter 2015."
Wilh. Wilhelmsen ASA. 2015. "Quarterly Report: Fourth Quarter 2015."
Wilh. Wilhelmsen ASA. 2015. "Quarterly Report: Second Quarter 2015."
Wilh. Wilhelmsen ASA. 2015. "Quarterly Report: Third Quarter 2015."
Wilh. Wilhelmsen ASA. 2016. "Annual Report."
Wilh. Wilhelmsen ASA. 2017. "Annual Report."
Wilh. Wilhelmsen ASA. 2018. "Annual Report."